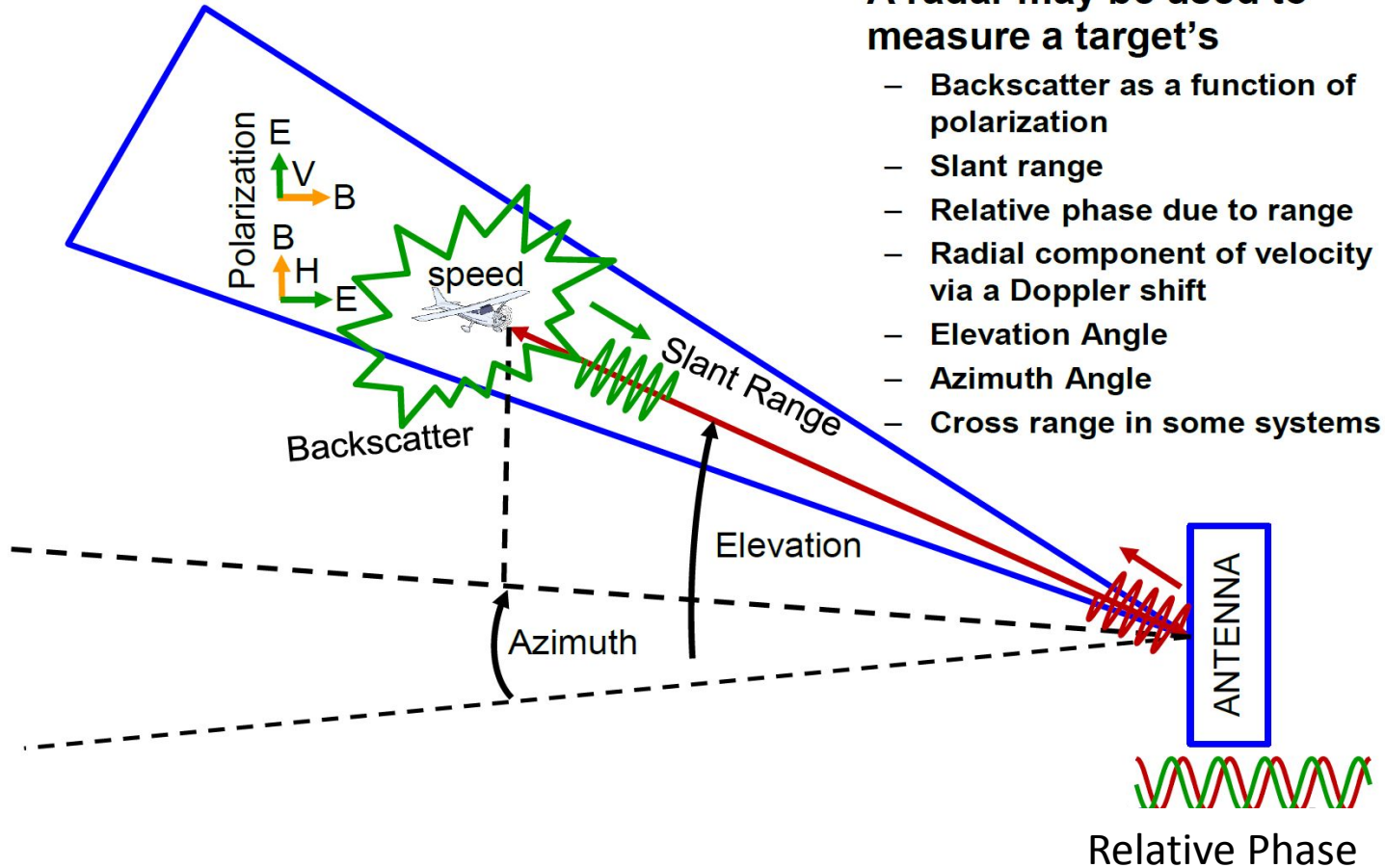


Radar Signal Models, part 1: Target and Clutter Mean RCS, Noise, and Doppler Shift

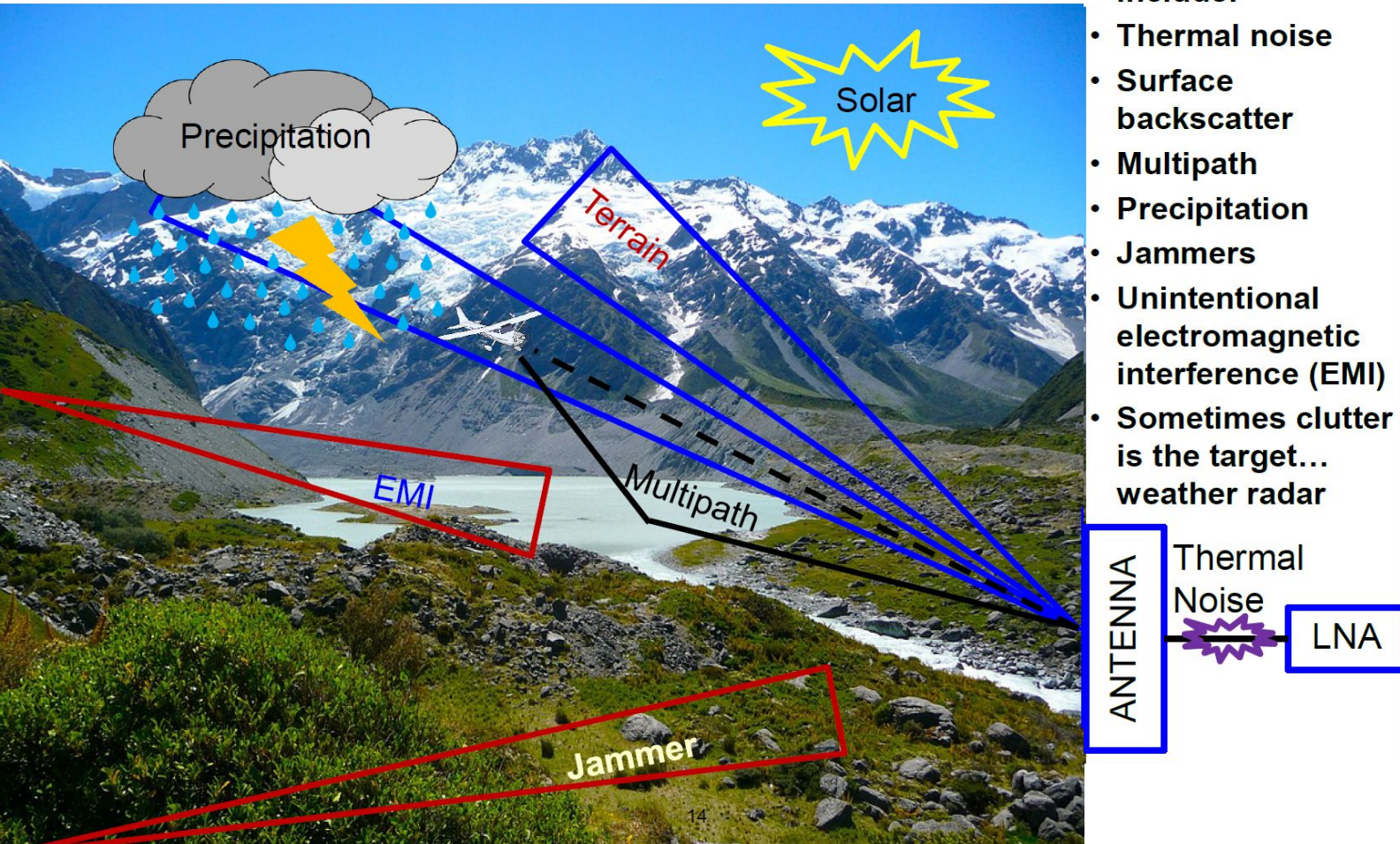
EE417 / EE517
CUA Spring 2024

Tuesdays 7:00-9:30pm
Instructor: Prof. Kevin Russo

What a Radar Measures.



Sources of Clutter or Interference



- Sources of clutter or interference include:
- Thermal noise
- Surface backscatter
- Multipath
- Precipitation
- Jammers
- Unintentional electromagnetic interference (EMI)
- Sometimes clutter is the target... weather radar

Radar Cross Section of Static Targets

The Baseband Signal

Real Signal

Transmitted pulse

$$\bar{x}(t) = A(t) \sin[2\pi F_t t + \theta(t)]$$

pulse envelope

carrier frequency

phase modulation

Complex Signal

**Received echo after coherent
demodulation to baseband**

$$y(t) = k \cdot A(t - t_0) \exp \left\{ j \left[-2\pi \frac{2R_0}{\lambda} + \theta(t - t_0) \right] \right\} + n(t)$$

delayed, scaled envelope

absolute phase offset

delayed phase modulation

noise

The important parameters to **measure** and **model** are:

- **Delay** t_0
- **Phase** $-2\pi \frac{2R_0}{\lambda} + \theta(t - t_0)$
- **Amplitude** $k \cdot |A|$ and its relation to power of noise n

Elements of the Radar Range Equation

- Radar radiates effective power $ERP = P_t G$
- Transmit power density reduces as $1/R^2$
- Target RCS has units of meters squared
 - Captures portion of signal and radiates toward radar
- Return signal power density reduces as $1/R^2$
- Receive antenna's effective area captures a portion of the signal $G\lambda^2$

$$A_{eff} = \frac{G\lambda^2}{4\pi}$$

Transmit
Effective
Radiated
Power

Units: Power/Area

$$\frac{P_t G}{4\pi R^2}$$

$$\frac{P_t G \sigma}{(4\pi)^2 R^4}$$

Units: Power/Area

$$P_{rec} = P_t G \frac{1}{4\pi R^2} \sigma \frac{1}{4\pi R^2} \frac{G\lambda^2}{4\pi} = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

Units: Power

Antenna Receive Area

Radar Cross Section

Units: Area

σ



$$\frac{1}{4\pi R^2}$$

Modeling Amplitude:

Radar Range Equation for a Point Target

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L_s L_a (R)} \quad \text{W}$$

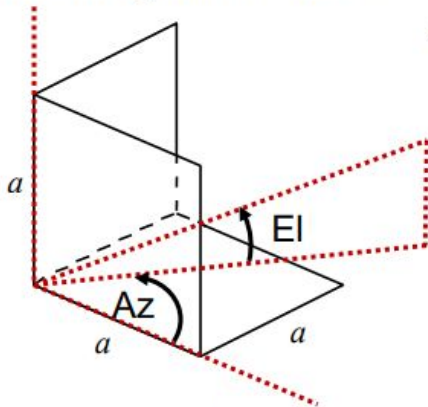
- **Signal power at the receiver input**
- **Signal amplitude is proportional to the square root of power**
- **For a point target, the received power decreases as R^4 , so**
 - **Doubling detection range requires**
 - A 16x (12 dB) transmitted power increase, *or*
 - A 4x (6 dB) antenna gain increase $\Rightarrow$ 4x antenna area increase
 - **Stealth: Halving detection range requires a 16x decrease in RCS σ**

P_r =	received power (W)
P_t =	transmitted power (W)
G =	antenna peak gain
λ =	wavelength (m)
σ =	target RCS (m ²)
R =	target range (m)
L_s =	system loss factor (dimensionless)
L_a =	atmospheric loss factor (dimensionless, function of range)

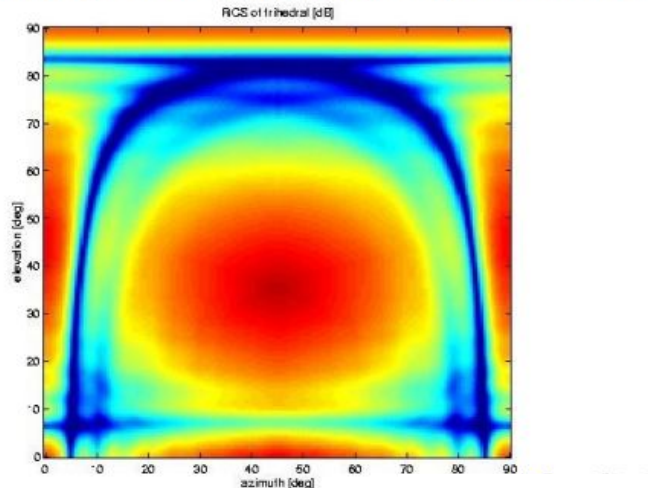
Radar Cross Section

- RCS is the **fictional** area over which the transmitted power density must be intercepted to collect a total power which, when re-radiated, would account for the receive power density
 - Depends on polarization, frequency, target shape, target aspect, target materials, ...
- Sphere of radius $a \gg \lambda$: $\sigma = \pi a^2$ Constant RCS (independent of frequency)

- Square trihedral of side length a : $\sigma = \frac{12\pi a^4}{\lambda^2}$



Aspect Angle and Frequency Dependent RCS



Typical Mean RCS at Microwave Frequencies

<i>Target</i>	<i>RCS, m²</i>	<i>RCS, dBsm</i>
Conventional unmanned winged missile	0.5	-3
Small single-engine aircraft	1	0
Small fighter aircraft or 4-passenger jet	2	3
Large fighter aircraft	6	8
Medium bomber or jet airliner	20	13
Large bomber or jet airliner	40	16
Jumbo jet	100	20
Small open boat	0.02	-17
Small pleasure boat	2	3
Cabin cruiser	10	10
Large ship at zero grazing angle	10,000+	40+
Bicycle	2	3
Automobile	100	20
Pickup truck	200	23
Insect	0.00001	-50
Bird	0.01	-20
Man	1	0

(After Skolnik, 2001)

Range Equation Example

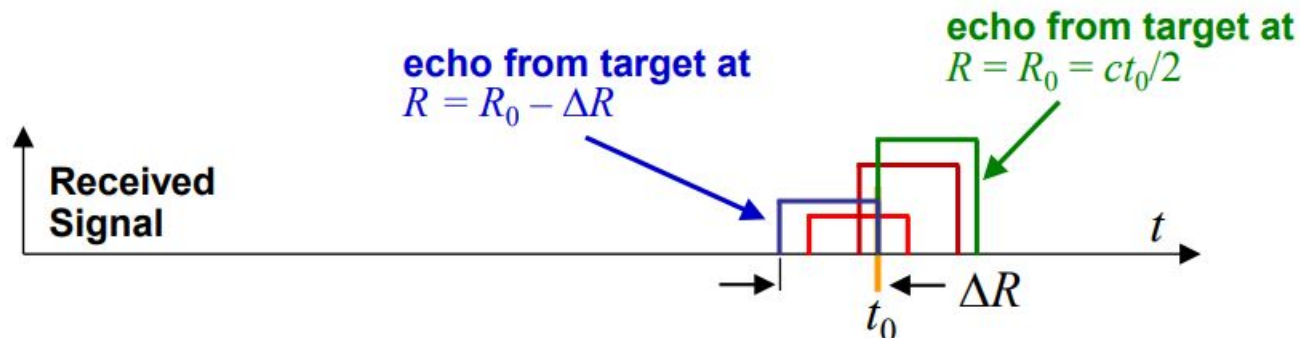
- **X band (10 GHz)**
 - $\lambda = 3 \times 10^8 / 10 \times 10^9 = 3 \text{ cm}$
- **Transmitted power $P_t = 1 \text{ kW} = +60 \text{ dBm}$**
- **Beamwidth (azimuth and elevation) = 1°**
 - $G \approx 20,000 / (1)(1) = 20,000 = 43 \text{ dB}$
- **Jumbo jet aircraft: RCS = 100 m^2**
- **Range $R = 10 \text{ km}$**
- **What is the received power P_r ?**

$$P_r = \frac{(1,000)(20,000)^2 (0.03)^2 (100)}{(4\pi)^3 (10,000)^4} = 1.814 \times 10^{-9} \text{ W}$$
$$= -57.4 \text{ dBm}$$

12 orders of magnitude!

RCS of Unresolved Scatterers - 1

- Consider a voltage measurement at a particular time t_0 with a pulse of length τ
 - Corresponding to range R_0 and range resolution $\Delta R = c\tau/2$
- Contributors to the measured voltage include
 - The leading edge of the echo from a scatterer at R_0
 - The trailing edge of the echo from a scatterer at $R_0 - \Delta R$
 - The middle of echoes from scatterers in between
- So scatterers within an interval of ΔR meters have overlapping echoes and are considered “unresolved”



RCS of Unresolved Scatterers - 2

- The contribution of scatterer i to the measured voltage is proportional to $\sqrt{\sigma_i} \exp[-j(4\pi/\lambda)R_i]$
- The total measured complex voltage is proportional to the sum $\sum_{i=1}^N \sqrt{\sigma_i} e^{-j(4\pi/\lambda)R_i}$ for all scatterers within a resolution cell
- So the measured power and RCS are proportional to $\left| \sum_{i=1}^N \sqrt{\sigma_i} e^{-j(4\pi/\lambda)R_i} \right|^2$
 - “Adding on a ‘voltage’ basis”
- The individual echo voltages combine **constructively and destructively** to form the measured voltage
- **The net RCS is very sensitive to relative phases!**

RCS of Unresolved Scatterers - 3

- The measured RCS of N unresolved scatterers is

$$\left| \sum_{i=1}^N \sqrt{\sigma_i} e^{-j(4\pi/\lambda)R_i} \right|^2$$

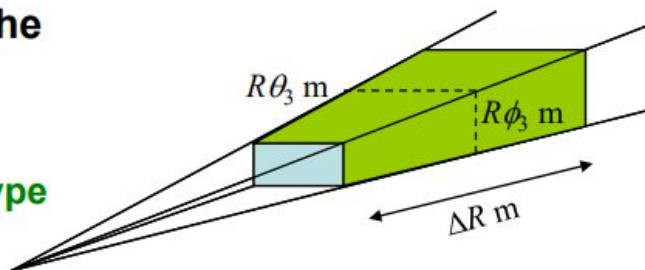
- “It can be shown” that, if the phases $-(4\pi/\lambda)R_i$ are completely random, then the **expected value** (mean, average) of the net RCS is the sum of the RCS’s of the individual scatterers:

$$\bar{\sigma} = \sum_{i=1}^N \sigma_i$$

- “**Adding on a ‘power’ basis**”
- Let’s use that fact to model clutter RCS ...

Volume Scattering RCS

- The observed mean RCS is modeled as the combined RCS of many unresolved scatterers uniformly distributed throughout the resolution cell
 - Rain, snow, clouds, chaff, insects, ...
- The mean RCS σ (dropping the overbar) is expressed as the product of two terms:
 - $\sigma = (\text{resolution cell volume in m}^3) \times (\text{reflectivity per unit volume of the type of clutter in the beam})$
- The RCS per cubic meter is called the **volume reflectivity η**
 - Units are $\text{m}^2/\text{m}^3 = \text{m}^{-1}$
 - η is a characteristic of the clutter type
- The resolution cell volume is approximately $[(\Delta R)(R\theta_3)(R\phi_3)]$
 - Resolution cell volume is a characteristic of the radar
 - θ_3 and ϕ_3 are the azimuth and elevation beamwidths (radians)
 - ΔR is the range resolution (m)

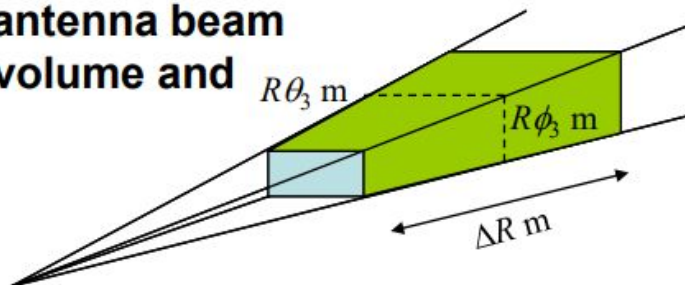


Range Equation for Volume Scattering

- Using $\sigma \approx [(\Delta R)(R \cdot \theta_3)(R \cdot \phi_3)]\eta$ from the previous page,

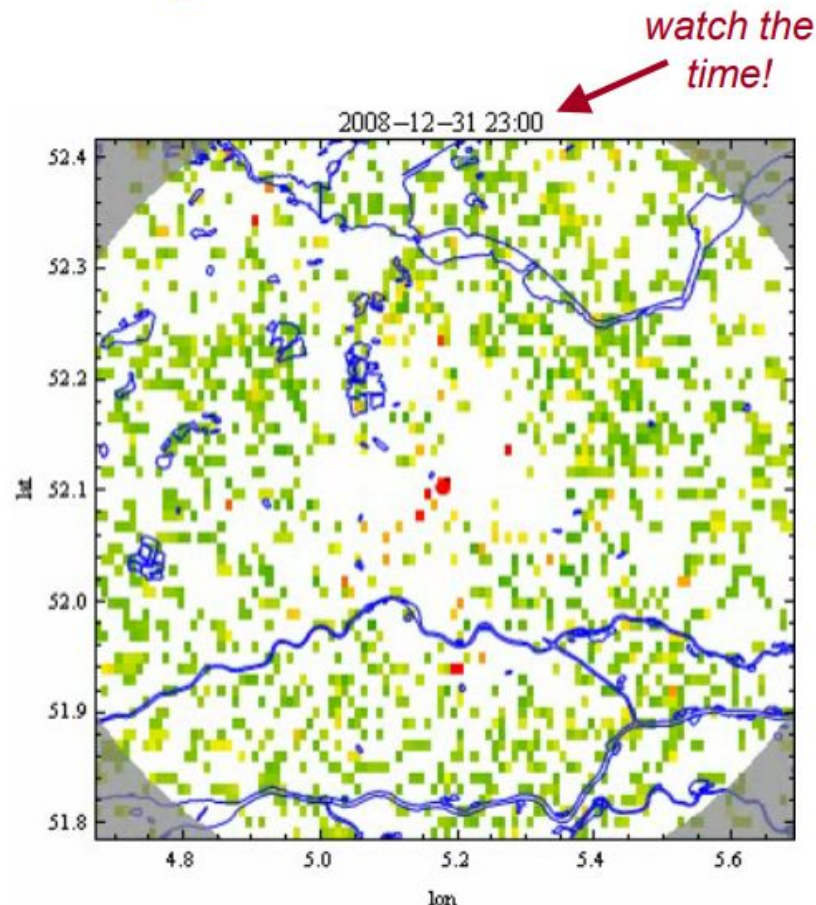
$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L_s L_a(R)} \xrightarrow{\sigma \approx [(\Delta R)(R \cdot \theta_3)(R \cdot \phi_3)]\eta} P_r = \frac{P_t G^2 \lambda^2 \eta \cdot \Delta R \cdot \theta_3 \cdot \phi_3}{(4\pi)^3 R^2 L_s L_a(R)}$$

- Note the received power decreases as R^2 , not R^4 as in point scatterer case
 - Because spreading of the antenna beam causes the resolution cell volume and therefore σ to grow as R^2



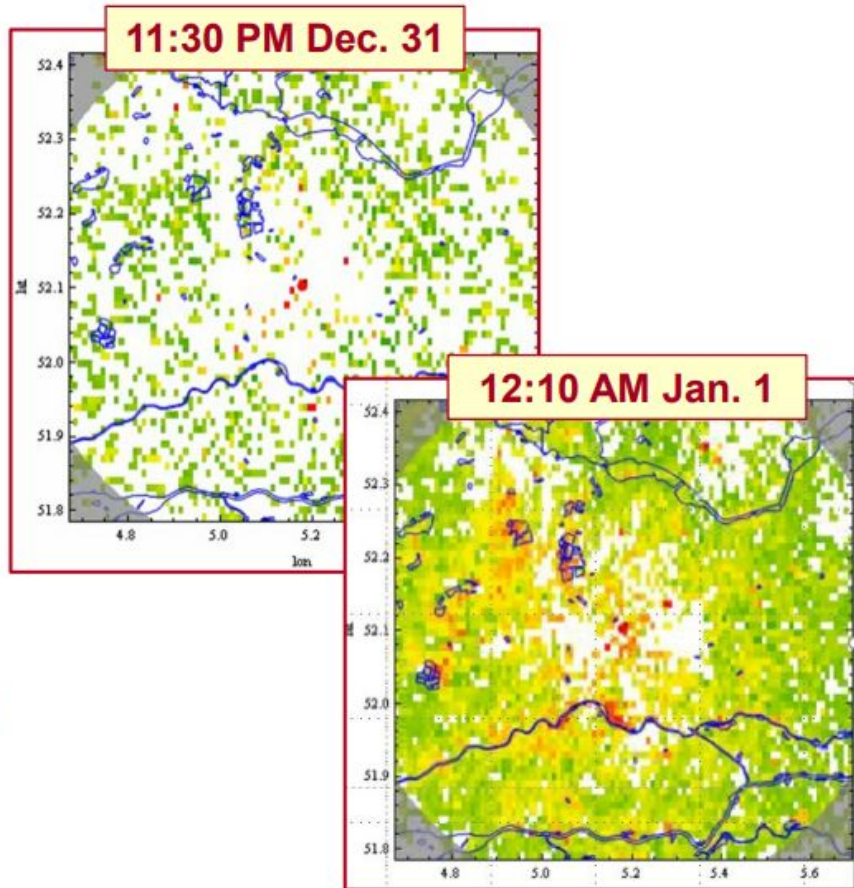
Volume Clutter Example: Bird Clutter

- Netherlands OPERA weather radar network, 2008-2009
- Birds take flight as New Year's fireworks are set off at midnight
- Forbes article:
<http://www.forbes.com/sites/grrlscientist/2015/12/31/birds-flee-en-mass-from-new-years-eve-fireworks/>
- Original source: Judy Shamoun-Baranes, Adriaan M. Dokter, Hans van Gasteren, E. Emiel van Loon, Hidde Leijnse, and Willem Bouten (2011). Birds flee en mass from New Year's Eve fireworks, *Behavioral Ecology*, 22:1173–1177 doi:[10.1093/beheco/arr102](https://doi.org/10.1093/beheco/arr102)



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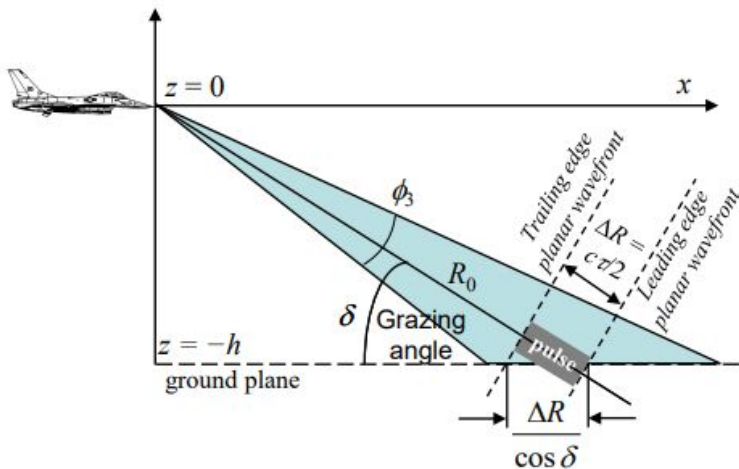
Area Scattering RCS

- The observed RCS is presumed to be due to scatterers uniformly distributed on a 2-D surface
 - Ground or sea clutter
- The mean RCS σ is expressed in terms of RCS per square meter, called *area reflectivity* or “*sigma-nought*” σ^0 :
 - Units are $\text{m}^2/\text{m}^2 = \text{dimensionless!}$
 - But often expressed in dB as $10\log_{10}(\sigma^0)$, called dBsm/sm
- The result is the point target range equation with
$$\sigma = (\text{resolution cell area}) \times (\text{reflectivity per unit area})$$
$$= (\text{resolution cell area}) \times \sigma^0$$
- *But there are two cases of resolution cell area to consider ...*

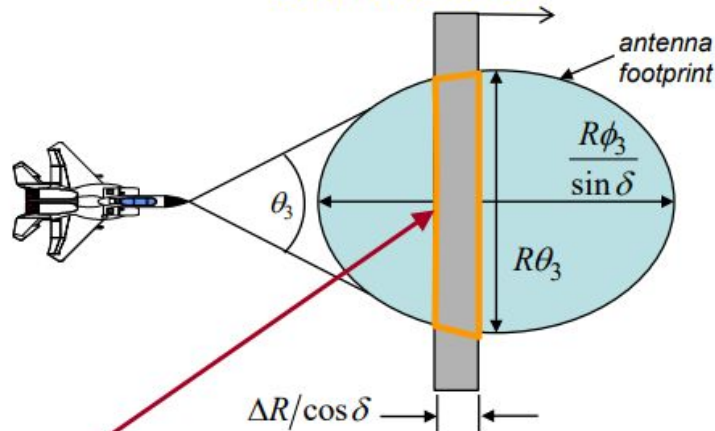
Clutter Patch Area: “Short” Pulse

- The range extent on the ground of the propagating pulse is shorter than that of the antenna elevation beam footprint
 - More likely at a shallow grazing angle δ

side view



overhead view



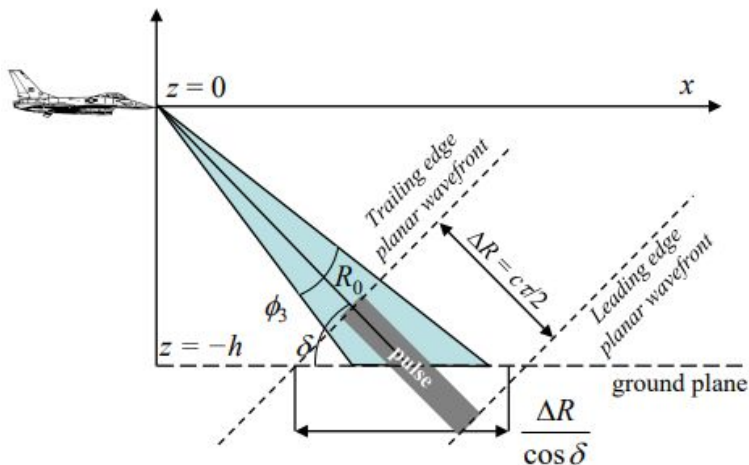
$$\text{Area} \approx \left(\frac{\Delta R}{\cos \delta} \right) (R\theta_3) \text{ at beam center}$$

- The effective clutter patch area is limited in the range dimension by the pulse footprint → “pulse-limited”

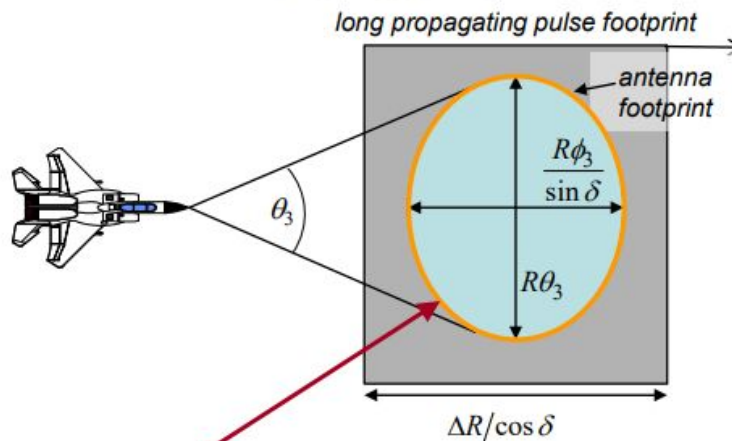
Clutter Patch Area: “Long” Pulse

- The range extent on the ground of the propagating pulse is longer than that of the antenna elevation beam footprint
 - More likely at a steep grazing angle δ

side view



overhead view



$$\text{Area} \approx \frac{\pi}{4} \left(\frac{R\phi_3}{\sin \delta} \right) (R\theta_3) \approx \left(\frac{R\phi_3}{\sin \delta} \right) (R\theta_3)$$

- The effective clutter patch area is limited in the range dimension by the elevation beam footprint → “beam-limited”

Determining Beam- vs. Pulse-Limited

- **Beam-limited** if the antenna elevation beam footprint limits the range extent of the contributing ground clutter:

$$\frac{R\phi_3}{\sin \delta} < \frac{\Delta R}{\cos \delta}$$

- **Pulse-limited** if the range resolution cell footprint limits the range extent of the contributing ground clutter:

$$\frac{R\phi_3}{\sin \delta} > \frac{\Delta R}{\cos \delta}$$

- **Therefore**

beam-limited:	$\frac{\Delta R}{R} \tan \delta > \phi_3$ (in radians!)
pulse-limited:	$\frac{\Delta R}{R} \tan \delta < \phi_3$ (in radians!)

- May transition from one to the other as range increases (and grazing angle decreases) or pulse length changes

Beam- and Pulse-Limited Range Equations

Beam-limited case:

- $\sigma = (\text{resolution cell area}) \times (\text{reflectivity per unit area})$
 $\approx [(R\phi_3/\sin\delta)(R\theta_3)]\sigma^0$

$$P_r = \frac{P_t G^2 \lambda^2 \sigma^0 \theta_3 \phi_3}{(4\pi)^3 R^2 L_s L_a (R) \sin \delta}$$

- **Received clutter power varies as R^{-2}**

Pulse-limited case:

- $\sigma = (\text{resolution cell area}) \times (\text{reflectivity per unit area})$
 $\approx [(\Delta R/\cos\delta)(R\theta_3)]\sigma^0$

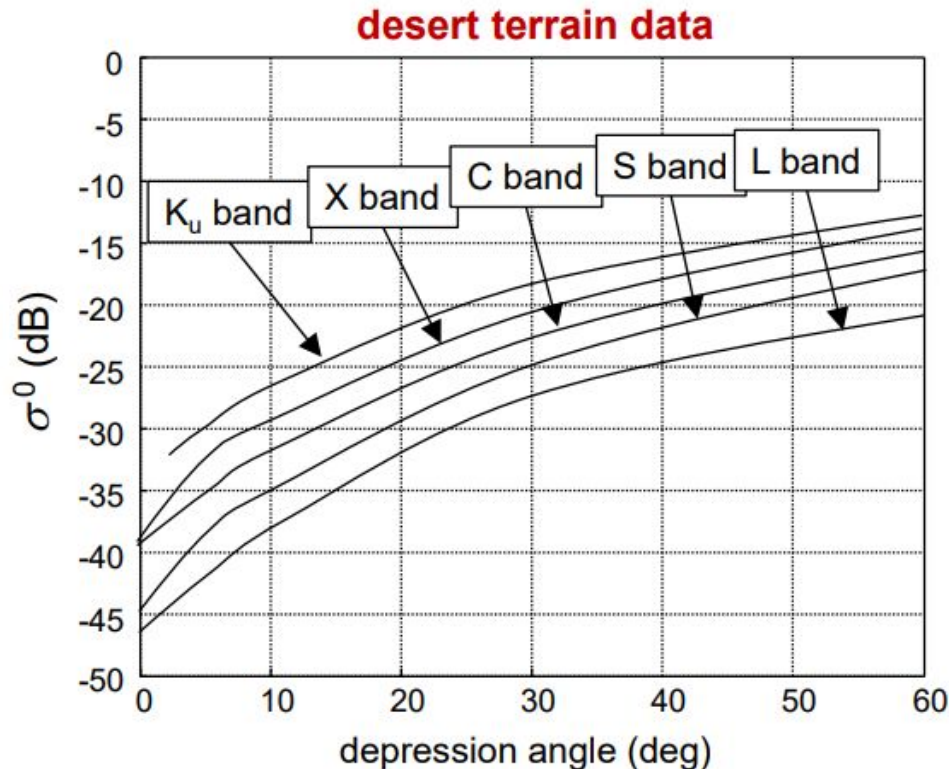
$$P_r = \frac{P_t G^2 \lambda^2 \sigma^0 \Delta R \theta_3}{(4\pi)^3 R^3 L_s L_a (R) \cos \delta}$$

- **Received clutter power varies as R^{-3}**

Note: Received clutter power will fall off faster than R^{-2} or R^{-3} if σ^0 changes with R , which is common at shallow grazing angles

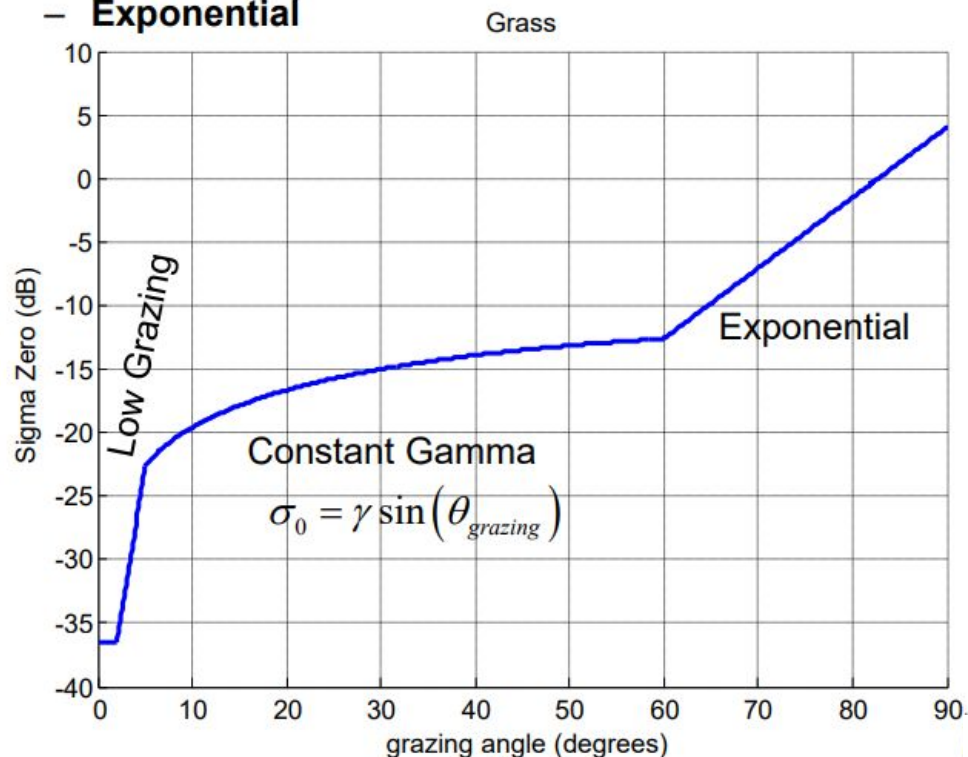
Example σ^0 : Desert Terrain

- Reflectivity varies with frequency and depression angle
- Also polarization, weather, ...
- And of course the type of material: desert, urban, forested, snow, calm seas, rough seas, ...



Another Example – Grass at C-Band

- Three commonly identified regions
 - Low grazing angle
 - Constant Gamma
 - Exponential



Doppler

The Doppler Phenomenon

- Frequency shift for an approaching target:

$$F_r = \left(\frac{1 + v/c}{1 - v/c} \right) F_t$$

- But v/c is always small (e.g., 2.2×10^{-6} for Mach 2), so

$$\begin{aligned} F_r &= (1 + v/c)(1 - v/c)^{-1} F_t \\ &= (1 + v/c) \left[1 + (v/c) + (v/c)^2 + \dots \right] F_t \\ &= \left[1 + 2(v/c) + 2(v/c)^2 + \dots \right] F_t \\ &\approx \left[1 + 2(v/c) \right] F_t = F_t + \underbrace{2v/\lambda}_{F_D \text{ Doppler shift}} \end{aligned}$$

Doppler Shift

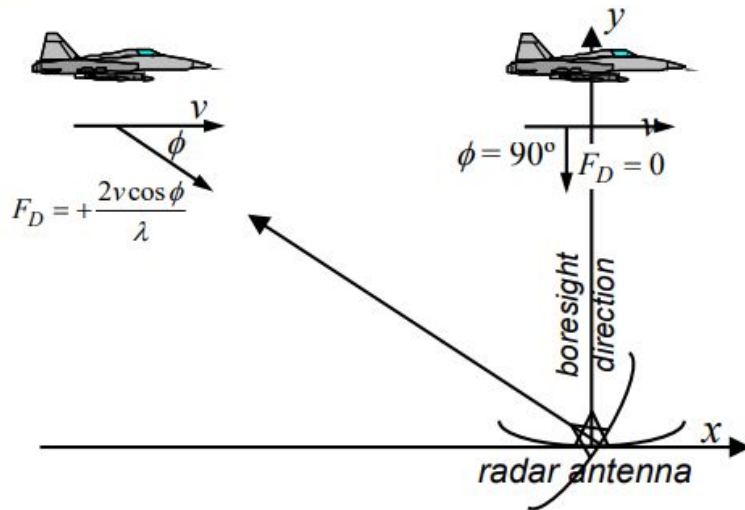
- Doppler **shift** is the **difference** between the original and shifted Doppler frequencies:

$$F_D = \frac{2v}{c} F = \frac{2v}{\lambda}$$

- It depends on the **radial** velocity between the radar and target

- It doesn't matter if the radar, target, or both are moving; just total radial velocity

- Doppler shift spreads the bandwidth also, but insignificantly



Band	Frequency (GHz)	Doppler shift (Hz) for $v = 1$ m/s
L	1	6.67
C	6	40.0
X	10	66.7
Ka	35	233
W	95	633

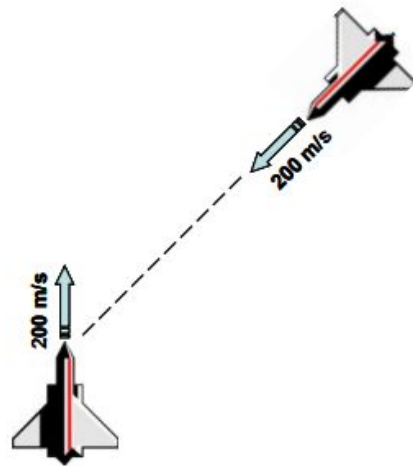
Example Doppler Shift

- An airborne radar is traveling north at 200 m/s. A target approaches from the NE, also traveling at 200 m/s. The radar is X band (10 GHz). What is the Doppler shift of the echo?

- Total closing velocity
 $= 200 + 200\cos(45^\circ)$
 $= 341.4 \text{ m/s}$

- $\lambda = c/F = 3 \times 10^8 / 1 \times 10^{10} = 0.03 \text{ m}$

- $F_D = 2v/\lambda = 2(341.4)/0.03 = 22.76 \text{ kHz}$



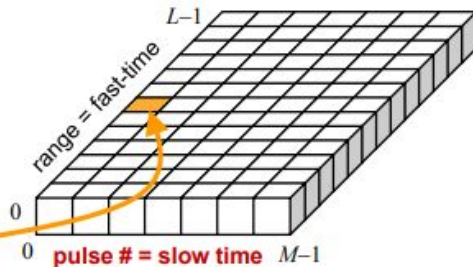
How to Measure Doppler - 1

- We need a long observation time to get good Doppler resolution
- We get this by using M pulses
 - So the total observation time is MT seconds
 - T is the pulse repetition interval (PRI)
- Remember that the phase of the echo is proportional to range:

$$\bar{y}(t) = A(t - 2R_0/c) e^{-j(4\pi/\lambda)R_0} e^{j2\pi Ft}$$

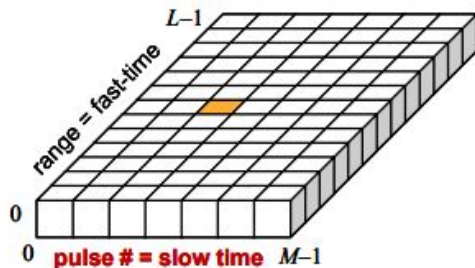
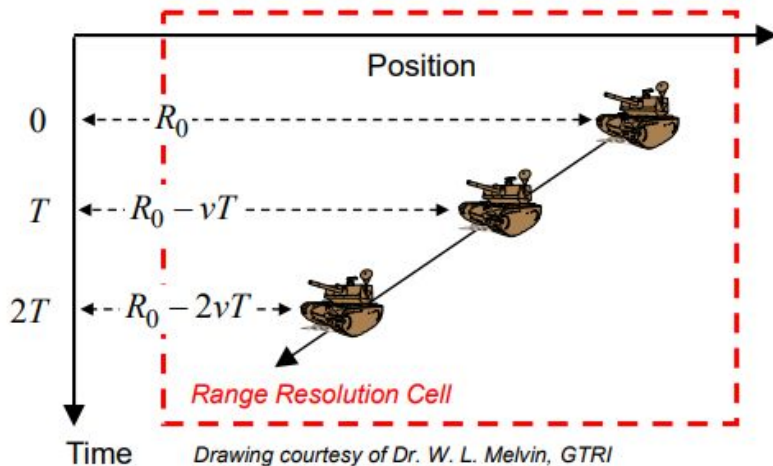
- Demodulate to remove the carrier, sample at time $t = 2R_0/c$ to get the baseband voltage sample for that range bin:

$$y(t) \Big|_{t=\frac{2R_0}{c}} = A e^{-j(4\pi/\lambda)R_0}$$



How to Measure Doppler - 2

- Consider a constant radial velocity target, so $R(t) = R_0 - vt$
 - Positive v corresponds to an approaching (“closing”) target
- On pulse #0, we measured $y[0] = Ae^{-j(4\pi/\lambda)R_0}$
- On pulse #1, the range will be $R_0 - vT \rightarrow$ we will measure $y[1] = Ae^{-j(4\pi/\lambda)(R_0 - vT)}$
- On pulse #2, the range will be $R_0 - 2vT \rightarrow$ we will measure $y[2] = Ae^{-j(4\pi/\lambda)(R_0 - 2vT)}$

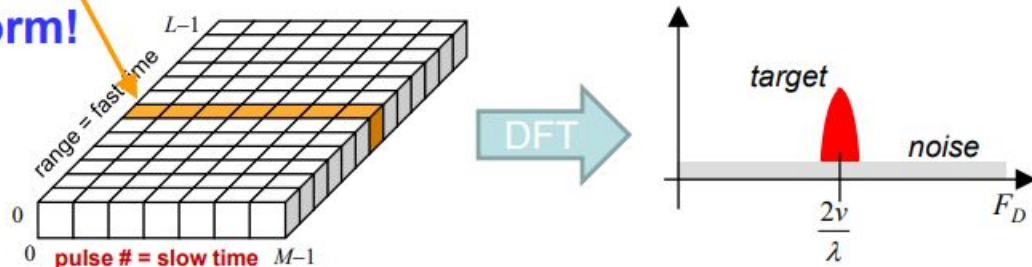


How to Measure Doppler - 3

- On pulse # m , the range will be $R_0 - mvT \rightarrow$ we will measure

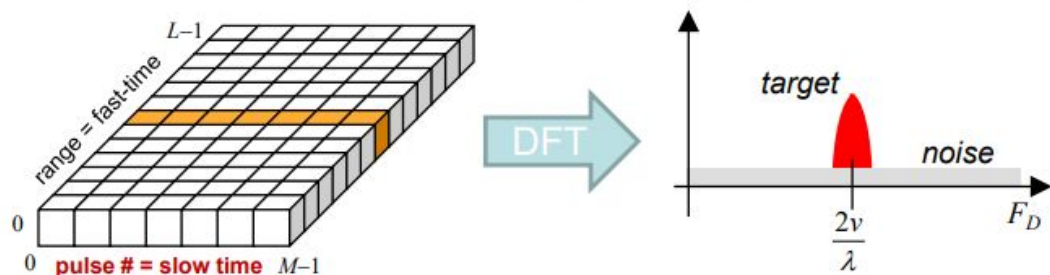
$$y[m] = Ae^{-j(4\pi/\lambda)(R_0 - mvT)} = \underbrace{Ae^{-j(4\pi/\lambda)R_0}}_{\text{constant}} \underbrace{e^{+j2\pi(2v/\lambda)mT}}_{\text{Doppler sinusoid}}$$

- The slow-time sequence of baseband samples forms a sampled sinusoid at the Doppler frequency!
- So, we can measure the Doppler shift with a Fourier transform!



Comments on Measuring Doppler

- Our model assumes that the target stays in the same range bin over the full CPI (M pulses) $\rightarrow$ no “range migration”
- Doppler shift, measured by the change in phase over multiple pulses in same range bin, is sometimes called **spatial Doppler**
 - We are really measuring range change vs. time via the pulse-to-pulse echo phase shifts
- This slow-time sequence of phase measurements is an example of the **phase history** of a target
 - Any constant radial velocity target will have a **linear** slow-time phase history
- The details of our math utilize the **stop-and-hop** assumption

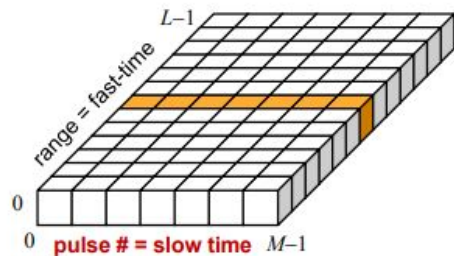


Comments on Range Walk or Migration

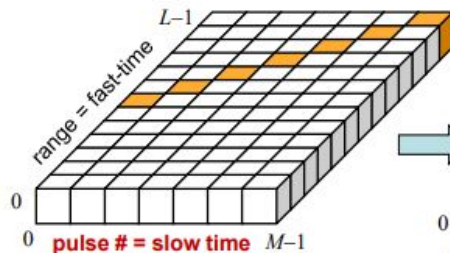
- Range walk may occur over a “long” CPI resulting in the target’s return moving across range cells
- In order to apply coherent processing across slow time, a range walk correction is required
- Range walk correction requires:
 - an estimate of the target’s radial velocity
 - or, in the case of SAR, the platform’s motion
- Range walk correction may be achieved using properties of the Fourier transform applied in fast time

$$x(t) \leftrightarrow X(\Omega)$$

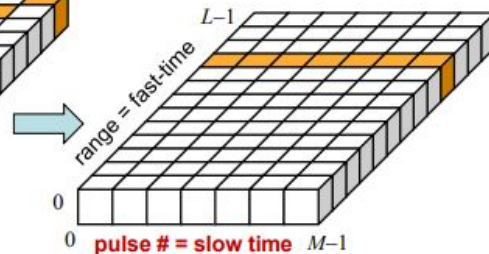
$$x(t - t_d) \leftrightarrow X(\Omega) \exp(-j\Omega t_d)$$



No Range Walk

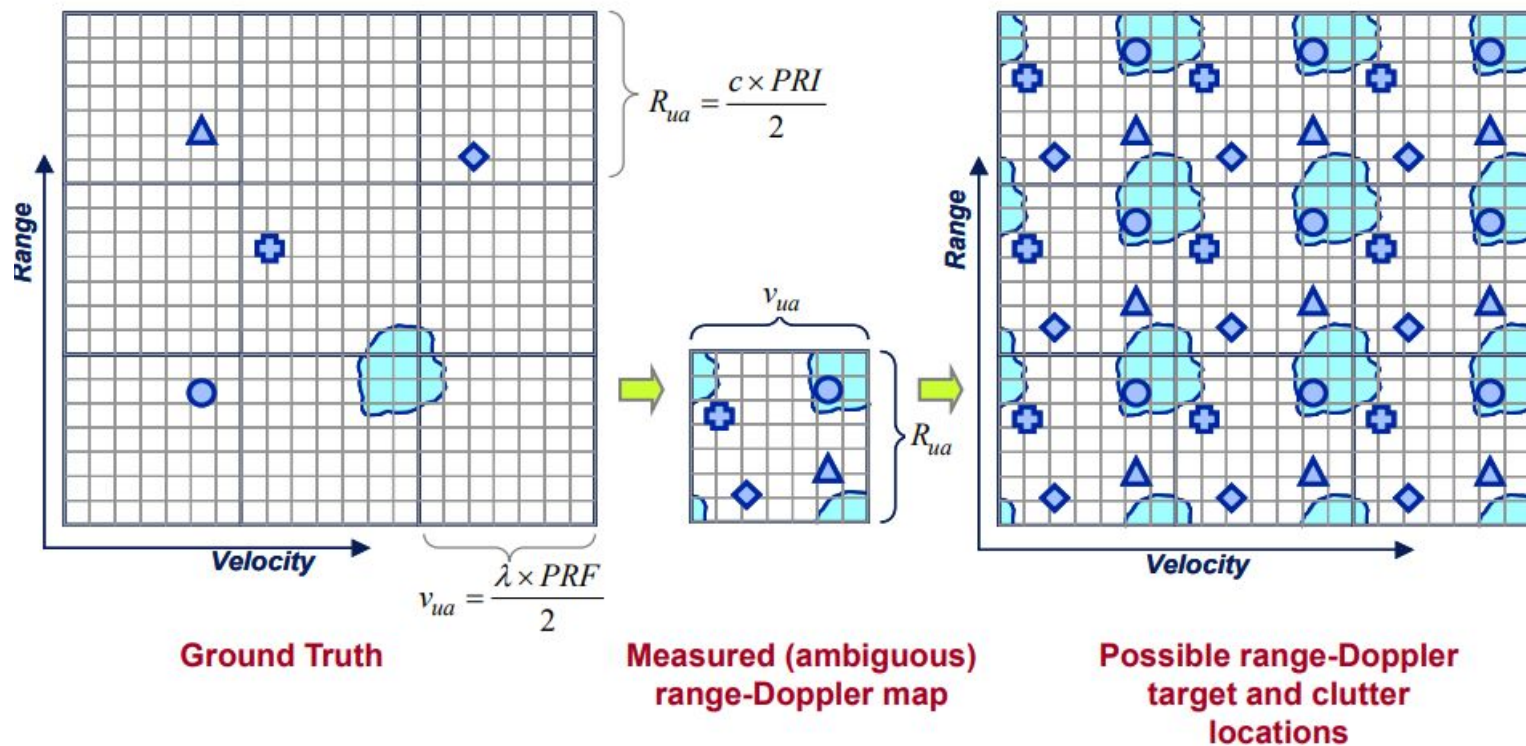


Range Walk



Range Walk Corrected

Effect of Range and Doppler Ambiguities on Range-Doppler Map



Noise

Thermal Noise

- The power spectrum of thermal noise:
$$S_n(F) = \frac{hF}{\exp[hF/kT] - 1} \text{ W/Hz}$$
 - T = temperature (°K)
 - k = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/°K}$
 - h = Planck's constant = $6.6254 \times 10^{-34} \text{ J-sec}$
- $F/T < 2 \times 10^9 \rightarrow hF/kT < 0.01 \rightarrow \exp(hF/kT) \approx 1 + hF/kT$
 $\rightarrow S_n(F) \approx kT \text{ W/Hz (a constant)}$
 - For instance, $F < 100 \text{ GHz}$ and $T > 50 \text{ °K}$
- Thus, the **noise power spectrum is white**
 - Equal power density at all frequencies
- This is the noise spectrum at the receiver **input**

Thermal Noise Power at Receiver Output

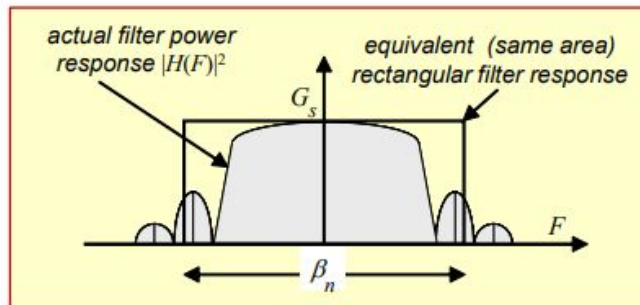
- Effect of a filter with frequency response $H(F)$ on the noise power spectrum:

$$S_{\text{out}}(F) = |H(F)|^2 S_{\text{in}}(F) \text{ W/Hz}$$

- The total filter output noise power is the integral of the power spectrum over frequency:

$$N_{\text{out}} = \int_{-\infty}^{\infty} |H(F)|^2 S_{\text{in}}(F) dF$$

$$= kT_0 \int_{-\infty}^{\infty} |H(F)|^2 dF \equiv kT_0 \cdot \beta_n G_s$$



- The total noise power at the receiver output is determined by
 - That portion of the white noise passed through noise-equivalent bandwidth β_n of the receiver, ...
 - Increased by the receiver gain G_s , and ...
 - Increased by the receiver noise figure F

$$F = \frac{SNR_{\text{IN}}}{SNR_{\text{OUT}}}$$

- Result:** $N_{\text{out}} = kT_0 \beta_n F G_s$

Output Signal-to-Noise Ratio

$$\begin{aligned}\text{SNR} \equiv \chi &= \frac{S_{\text{out}}}{N_{\text{out}}} = \frac{G_s P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L_s L_a(R)} \cdot \frac{1}{kT_0 \beta_n F G_s} \\ &= \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 kT_0 \beta_n F L_s L_a(R)}\end{aligned}$$

- This is just our earlier range equation for received signal power, with $kT_0 \beta_n F$ added to the denominator
 - The receiver gain G_s cancels out
- Similar results for volume and area clutter are obtained by adding $kT_0 \beta_n F$ to the denominators of alternate range equations